# DIABETES PREDICTION

## CAPSTONE PROJECT TECHNICAL PRESENTATION

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POST GRADUATE DIPLOMA IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING - JUNE 2025

JULY 17, 2026

# DATA

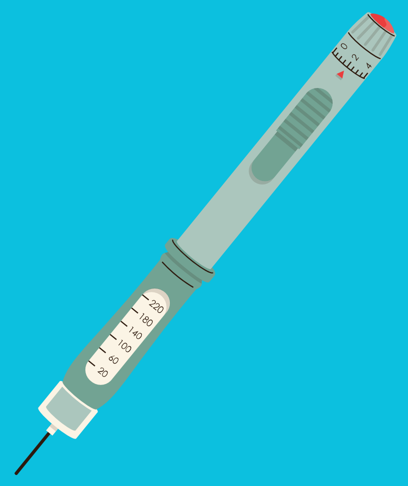
kaggle

Pima Indians Diabetes Dataset

768 FEMALE PATIENTS

AT LEAST 21 YEARS OLD

8 FEATURES PLUS OUTCOME



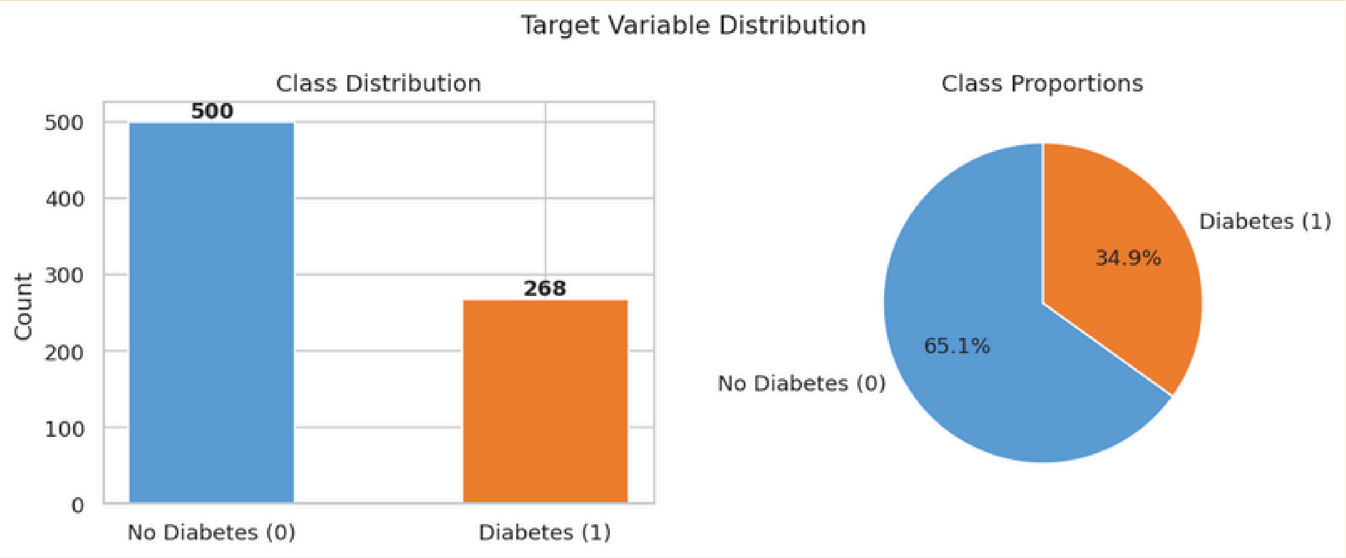
# DATA DICTIONARY

| Feature | Type | Description | Units / Range |
|---|---|---|---|
| Pregnancies | int | Number of times pregnant | 0–17 |
| Glucose | int | Plasma glucose concentration (2hr oral glucose tolerance test) | mg/dL |
| BloodPressure | float | Diastolic blood pressure | mm Hg |
| SkinThickness | int | Triceps skinfold thickness | mm |
| Insulin | float | 2-hour serum insulin | μU/mL |
| BMI | float | Body mass index | $kg/m^2$ |
| DiabetesPedigreeFunction | float | Genetic risk score based on family history | 0.0–2.5 |
| Age | int | Patient age | Years (21–81) |
| Outcome | int | Target variable | 0 = No Diabetes, 1 = Diabetes |

# DATA PREPROCESSING, APPLIED EDA & FEATURE ENGINEERING

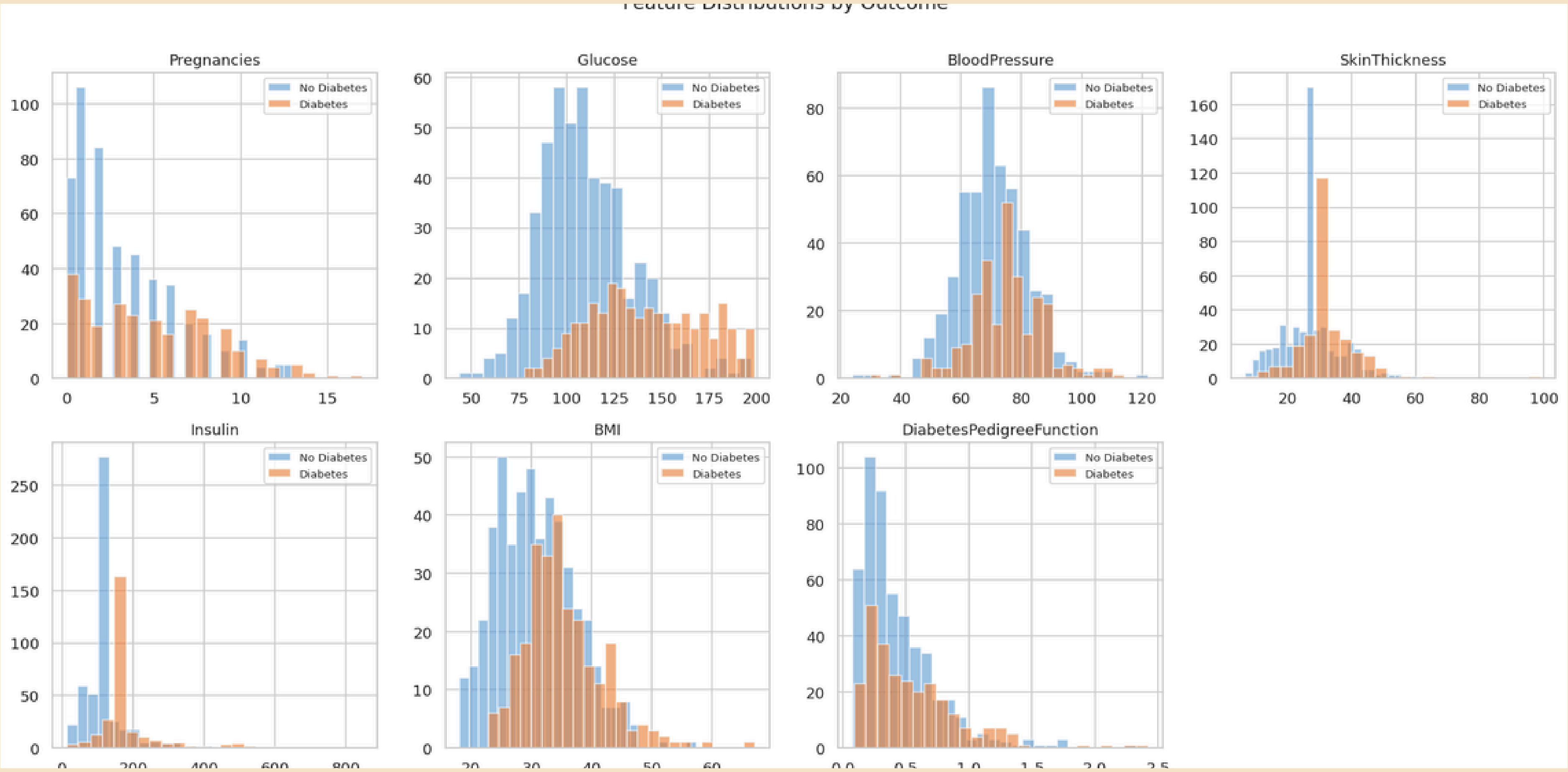
CHECKED FOR MISSING VALUES ✅

CLASS BALANCE ANALYSIS ✅



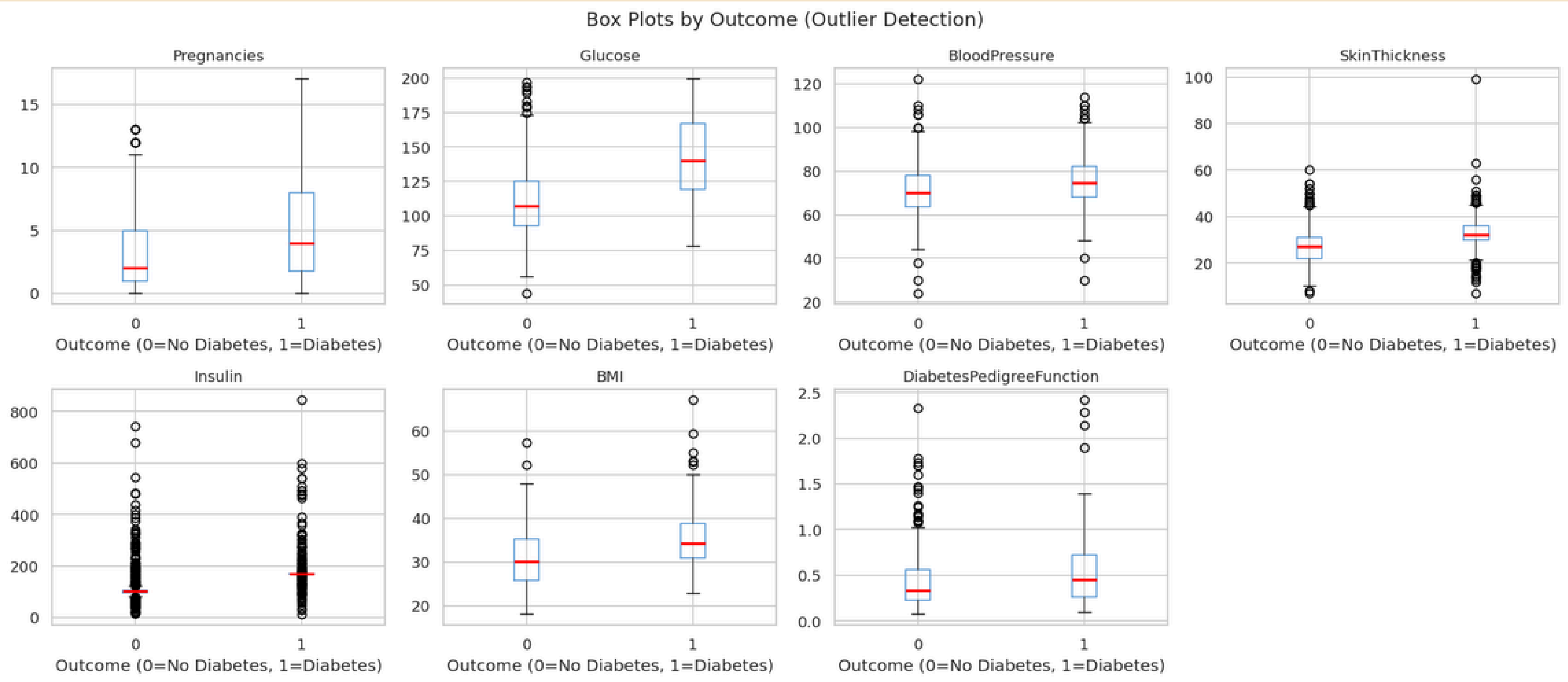

Class imbalance ratio: 1.87:1 (majority:minority) ⚠️
Imbalanced dataset – use class_weight="balanced" or SMOTE when training models.

FEATURE DISTRIBUTION ✅



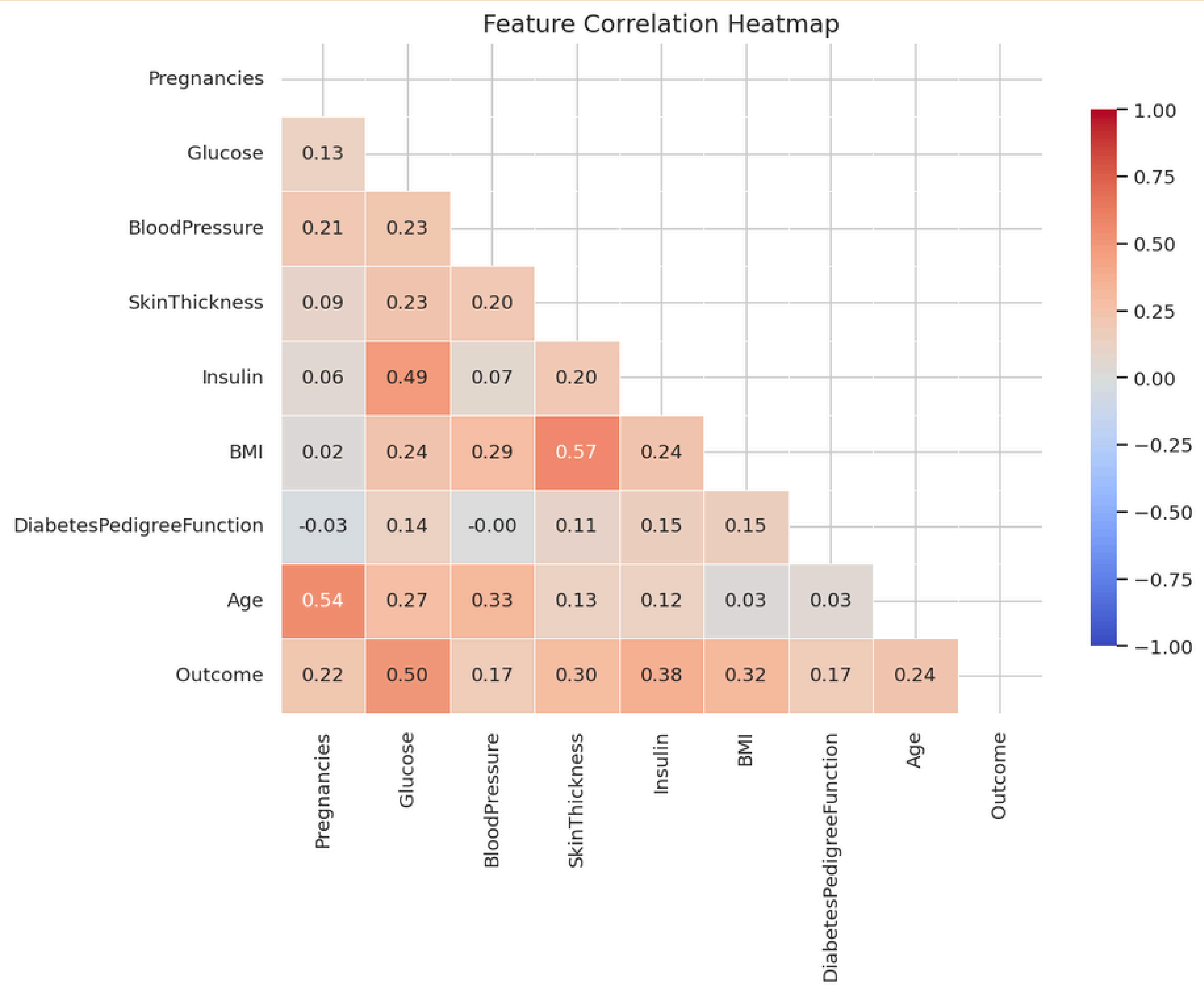
# DATA PREPROCESSING, APPLIED EDA & FEATURE ENGINEERING

## OUTLIER DETECTION ✅




Insulin is the noisiest feature with the most outiers.

## CORRELATION HEATMAP ✅




Glucose is the strongest predictor.

# DATA PREPROCESSING, APPLIED EDA & FEATURE ENGINEERING

## FEATURE ENGINEERING ✅

| index | AgeGroup | BMICategory | GlucoseRisk | InsulinGlucoseRatio | HighRisk |
|---|---|---|---|---|---|
| 0 | 41-50 | Obese | Diabetic | 1.13758389261745 | 1 |
| 1 | 31-40 | Overweight | Normal | 1.19186046511628 | 0 |
| 2 | 31-40 | Normal | Diabetic | 0.921195652173913 | 0 |
| 3 | 21-30 | Overweight | Normal | 1.04444444444444 | 0 |
| 4 | 31-40 | Obese | Diabetic | 1.21739130434783 | 0 |

High risk if they simultaneously have Glucose over 140, BMI over 30, AND Age over 40

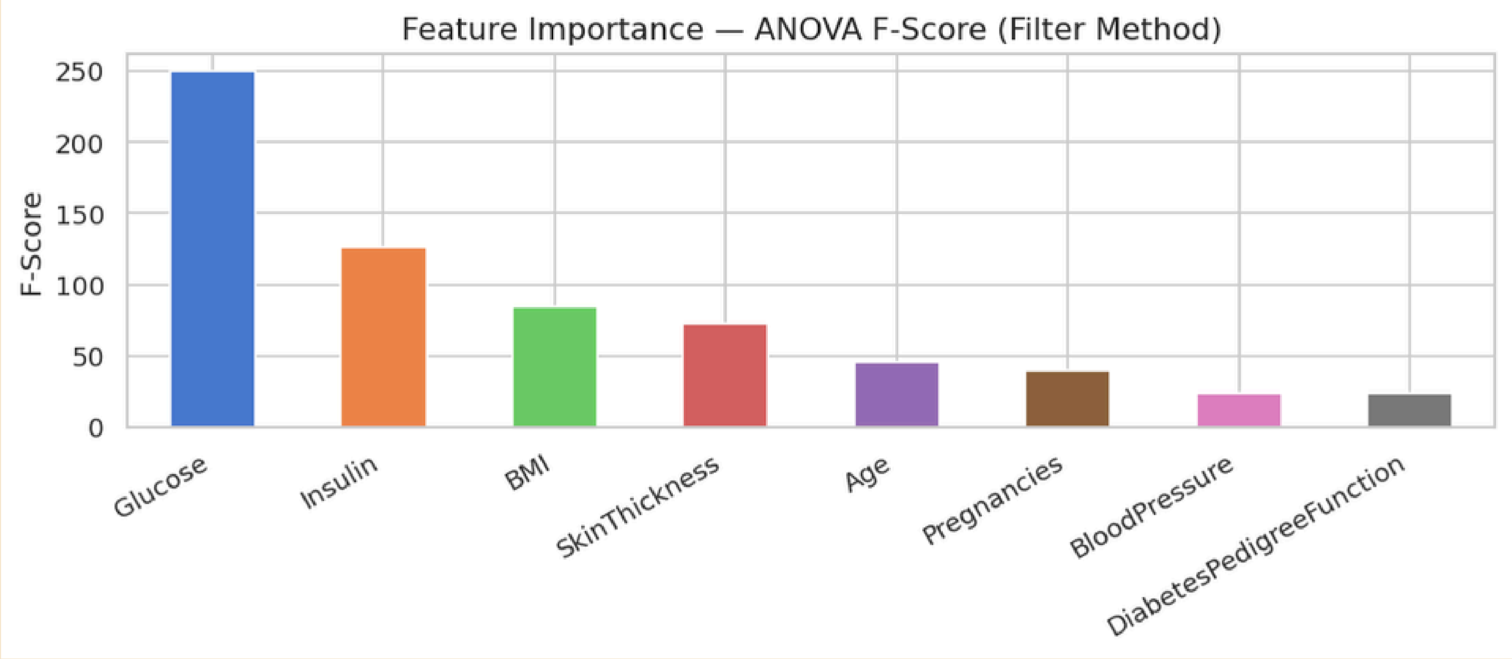
## FEATURE SCALING ✅

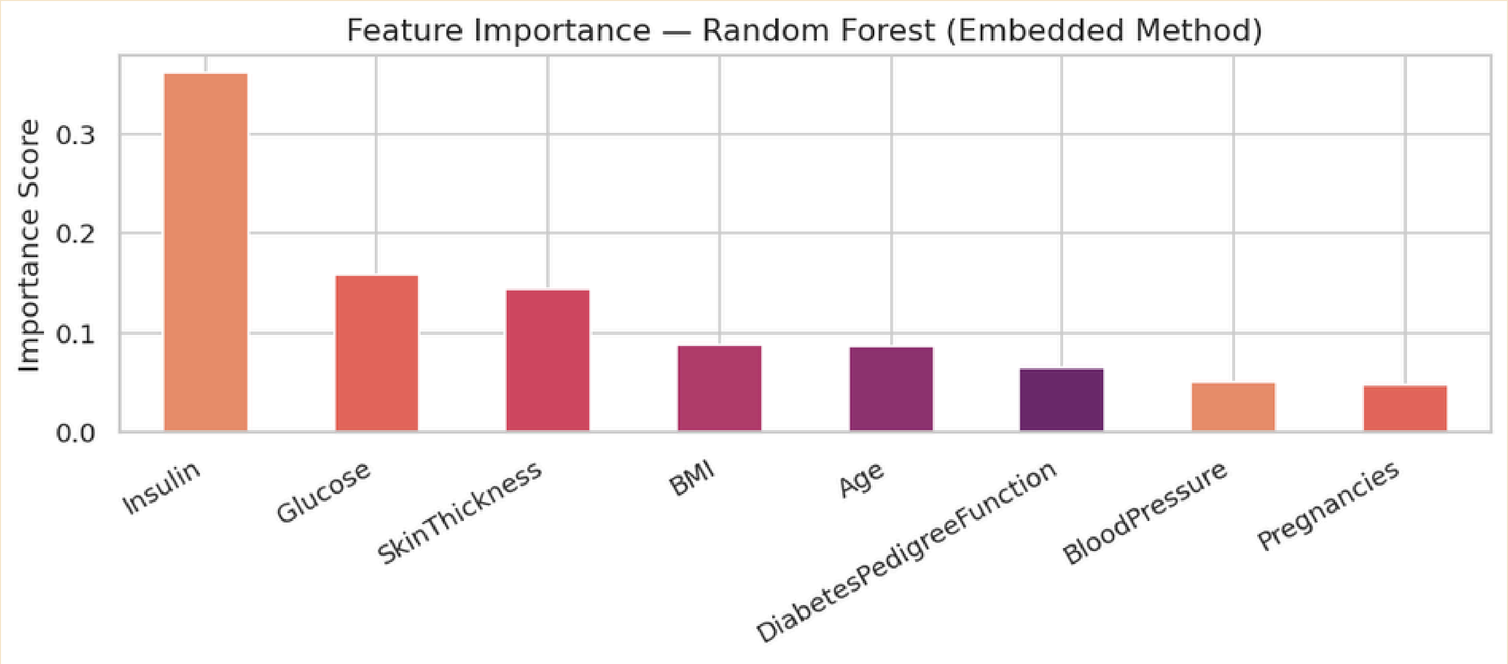
| | Pregnancies | Glucose | Blood Pressure | Skin Thickness | Insulin | BMI | Diabetes Pedigree Function | Age |
|---|---|---|---|---|---|---|---|---|
| **count** | 768 | 768 | 768 | 768 | 768 | 768 | 768 | 768 |
| **mean** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| **std** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| **min** | -1.14 | -2.55 | -4 | -2.49 | -1.43 | -2.07 | -1.19 | -1.04 |
| **25%** | -0.84 | -0.72 | -0.69 | -0.46 | -0.44 | -0.72 | -0.69 | -0.79 |
| **50%** | -0.25 | -0.15 | -0.03 | -0.12 | -0.44 | -0.06 | -0.3 | -0.36 |
| **75%** | 0.64 | 0.61 | 0.63 | 0.33 | 0.31 | 0.61 | 0.47 | 0.66 |
| **max** | 3.91 | 2.54 | 4.1 | 7.87 | 7.91 | 5.04 | 5.88 | 4.06 |

Method used - Standard Scaler

# DATA PREPROCESSING, APPLIED EDA & FEATURE ENGINEERING

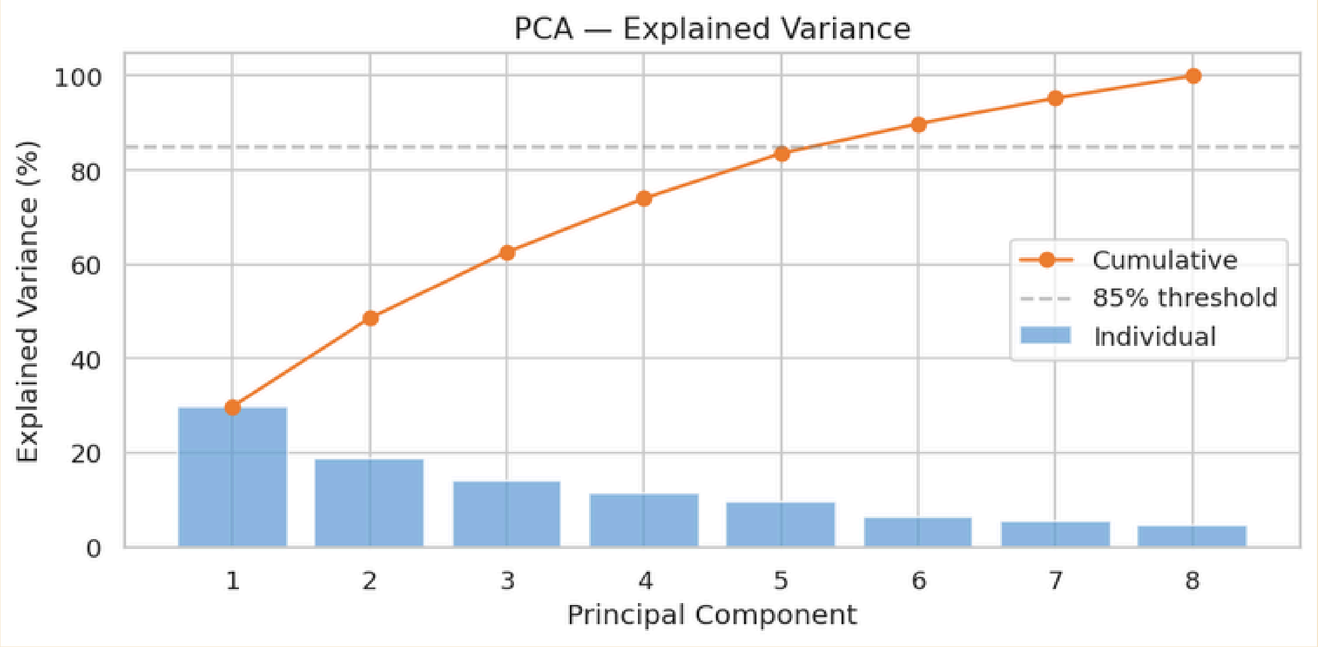
## FEATURE SELECTION ✅

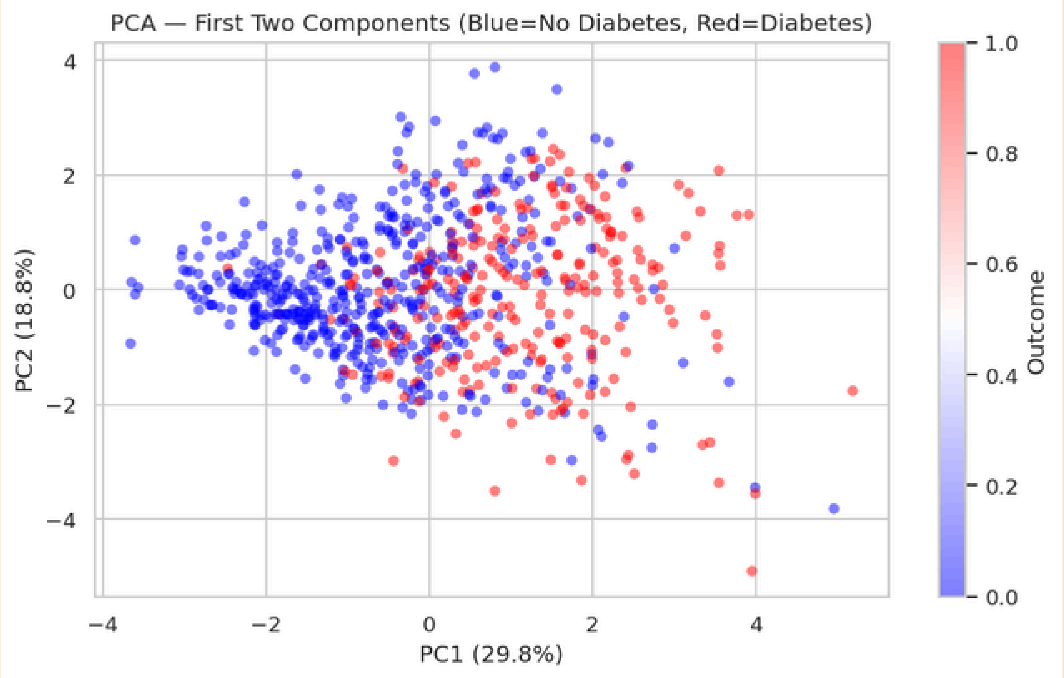



Both methods agree that the top two features are Glucose and Insulin.

## DIMENSIONALITY REDUCTON - PCA ✅




8 features can be compressed to 5 and explain 85% of the variance.

# MODEL IMPLEMENTATION

## TEST / TRAIN SPLIT & SMOTE

ORIGINAL 768 patients (80/20 split)
├ Training Set (614) → SMOTE applied → 800 balanced patients
~400 No Diabetes 400 No Diabetes
~214 Diabetes +186 synthetic → 400 Diabetes

└Test Set (154) → NO SMOTE → stays at 154 real patients
~100 No Diabetes (natural imbalance preserved)
~54 Diabetes

## MODEL COMPARISON

MODEL COMPARISON (sorted by AUC-ROC)

| | Accuracy | AUC-ROC | F1-Score | Precision | Recall |
|---|---|---|---|---|---|
| XGBoost | 0.8766 | 0.9467 | 0.8288 | 0.8070 | 0.8519 |
| Random Forest | 0.8766 | 0.9447 | 0.8257 | 0.8182 | 0.8333 |
| Decision Tree | 0.8636 | 0.8944 | 0.8142 | 0.7797 | 0.8519 |
| Logistic Regression | 0.7468 | 0.8161 | 0.6929 | 0.6027 | 0.8148 |

## MODEL TRAINING

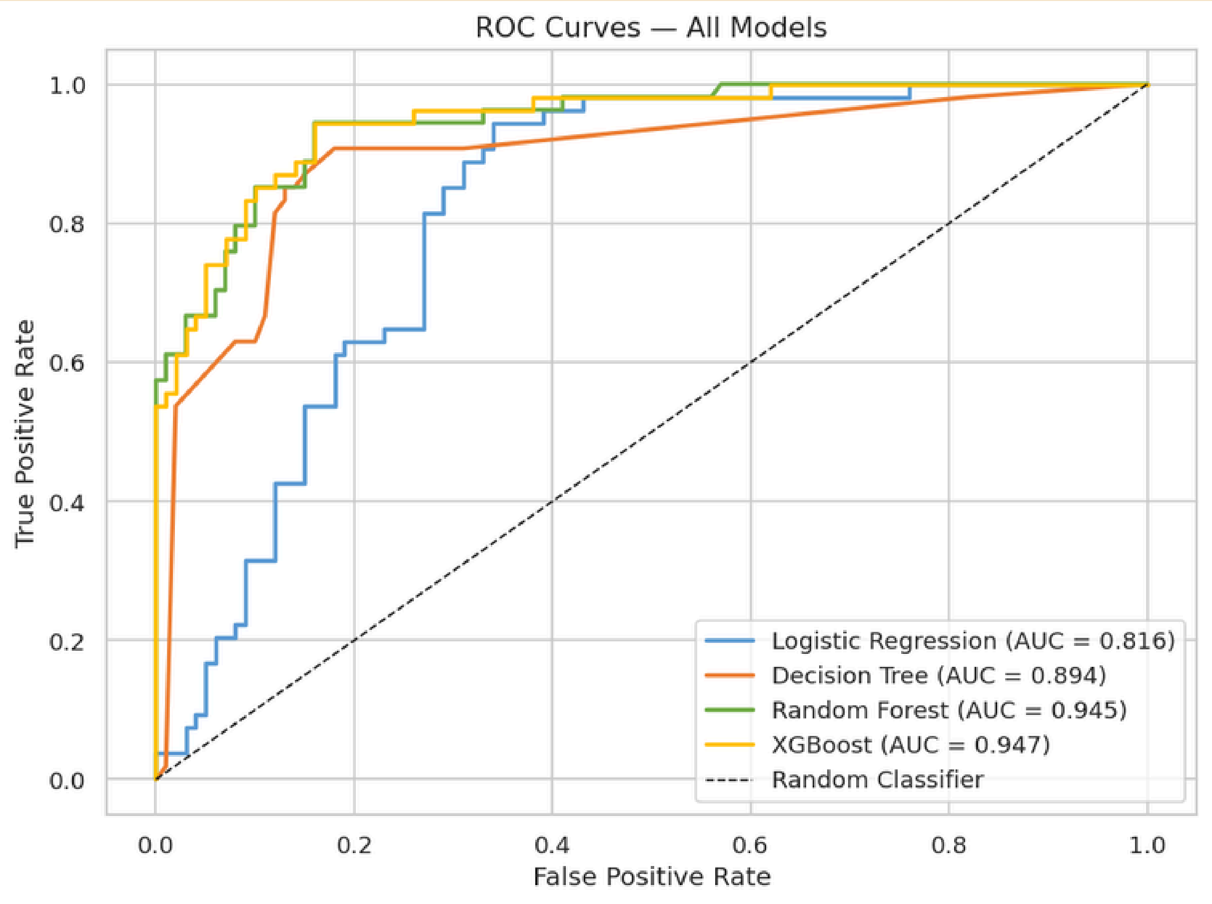
4 MODELS

1. Logistic Regression – interpretable baseline
2. Decision Tree – visual, rule-based
3. Random Forest – ensemble of trees, reduces variance
4. XGBoost – gradient boosting, typically best performer

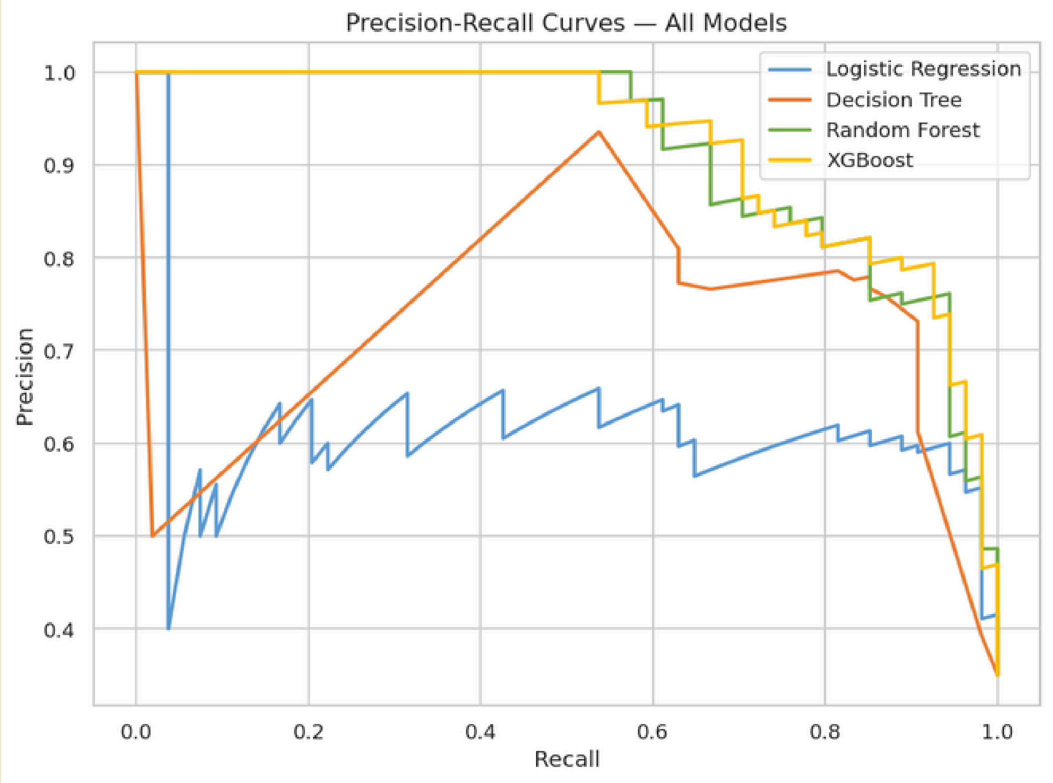
## MODEL COMPARISON

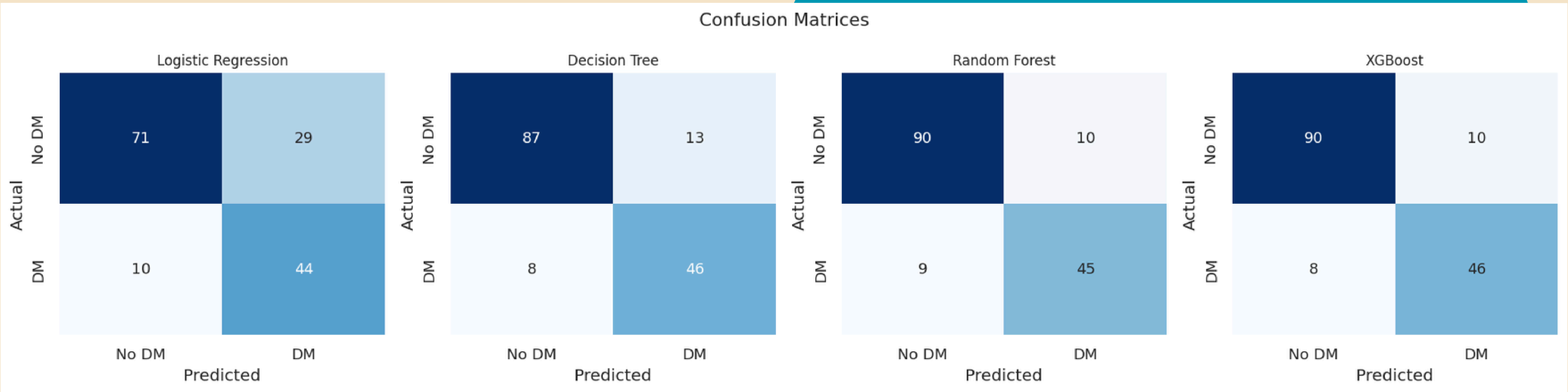
# MODEL IMPLEMENTATION

## ROC CURVES




## PRECISION-RECALL CURVES

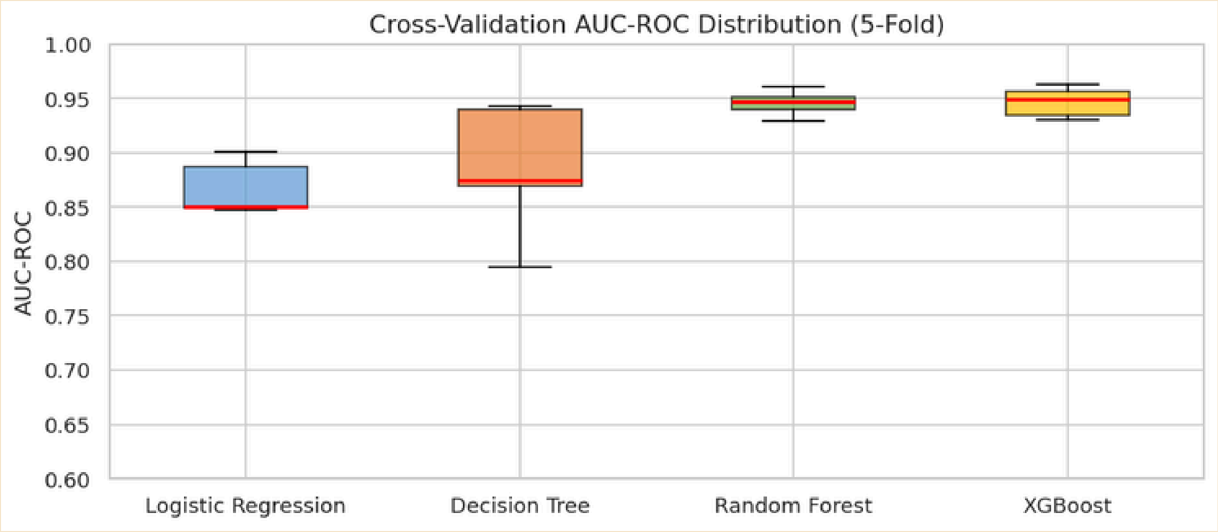



## CONFUSION MATRICES

# MODEL IMPLEMENTATION

## CROSS VALIDATION




5-Fold Stratified Cross-Validation AUC-ROC:

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| | |
|---|---|
| Logistic Regression | : 0.8669 ± 0.0226 |
| Decision Tree | : 0.8843 ± 0.0545 |
| Random Forest | : 0.9458 ± 0.0108 |
| XGBoost | : 0.9458 ± 0.0107 |

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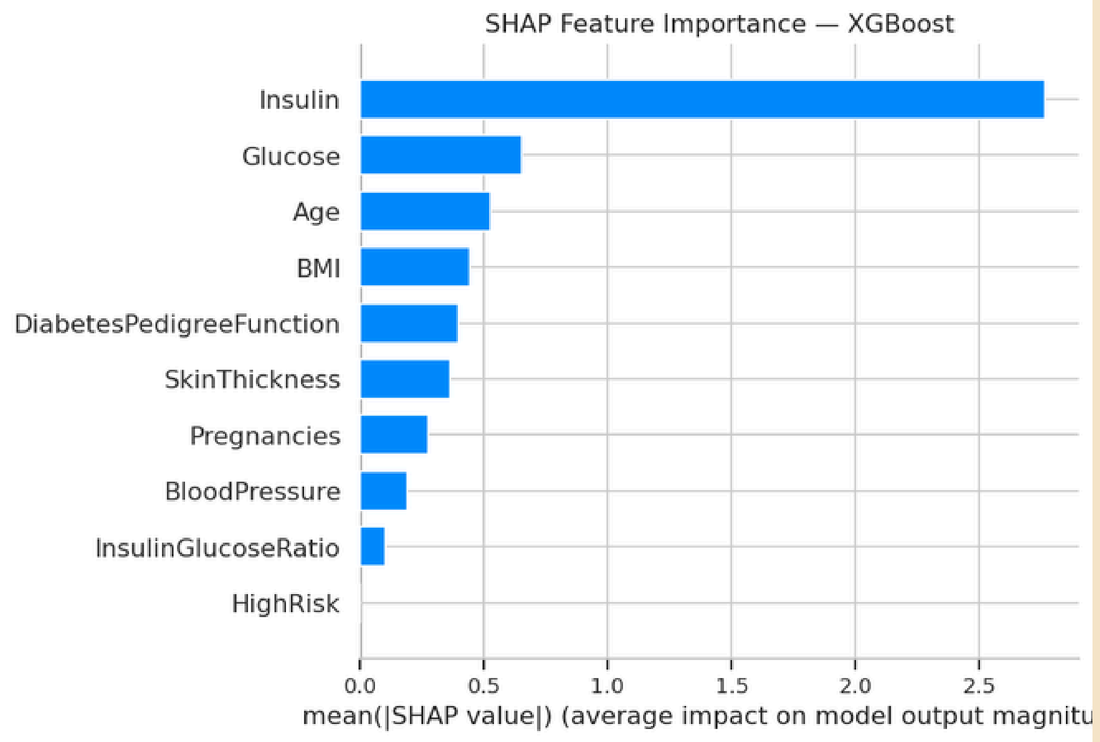
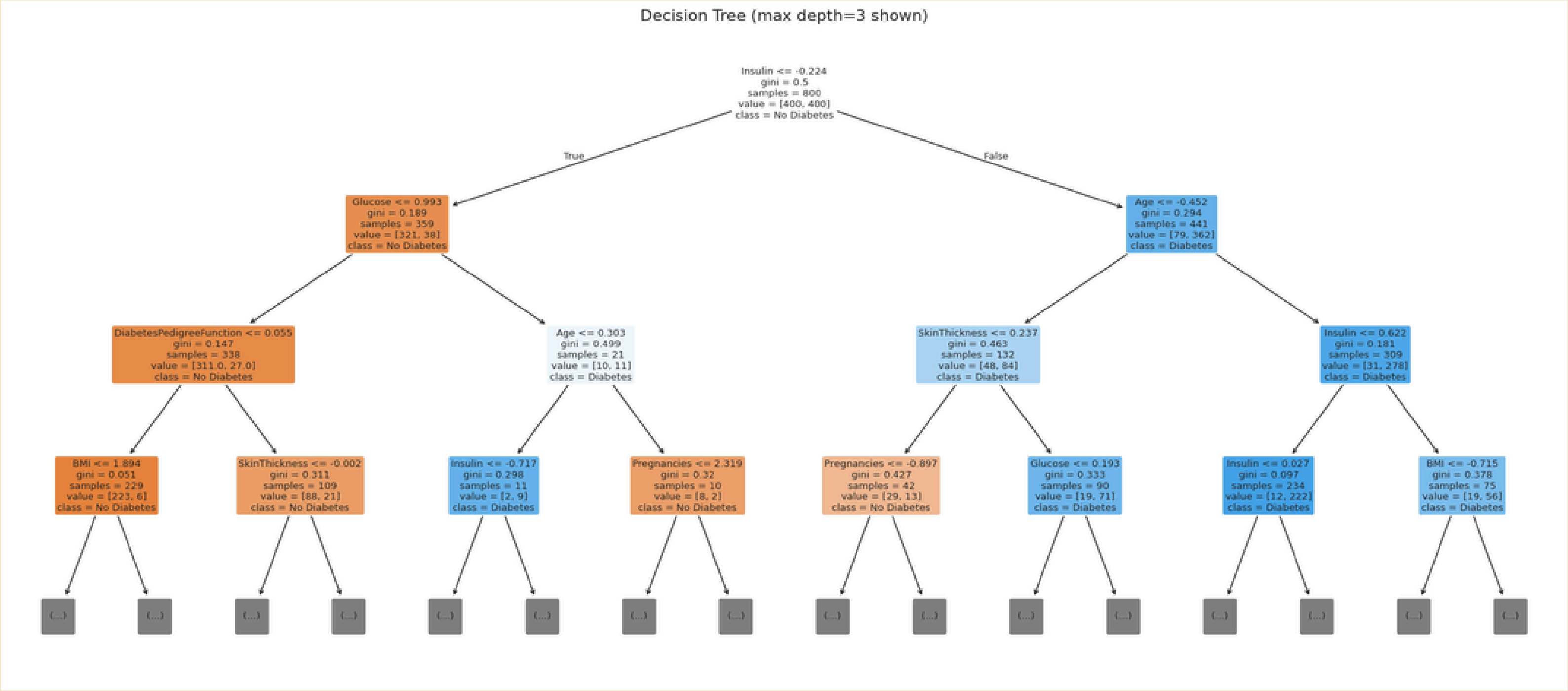
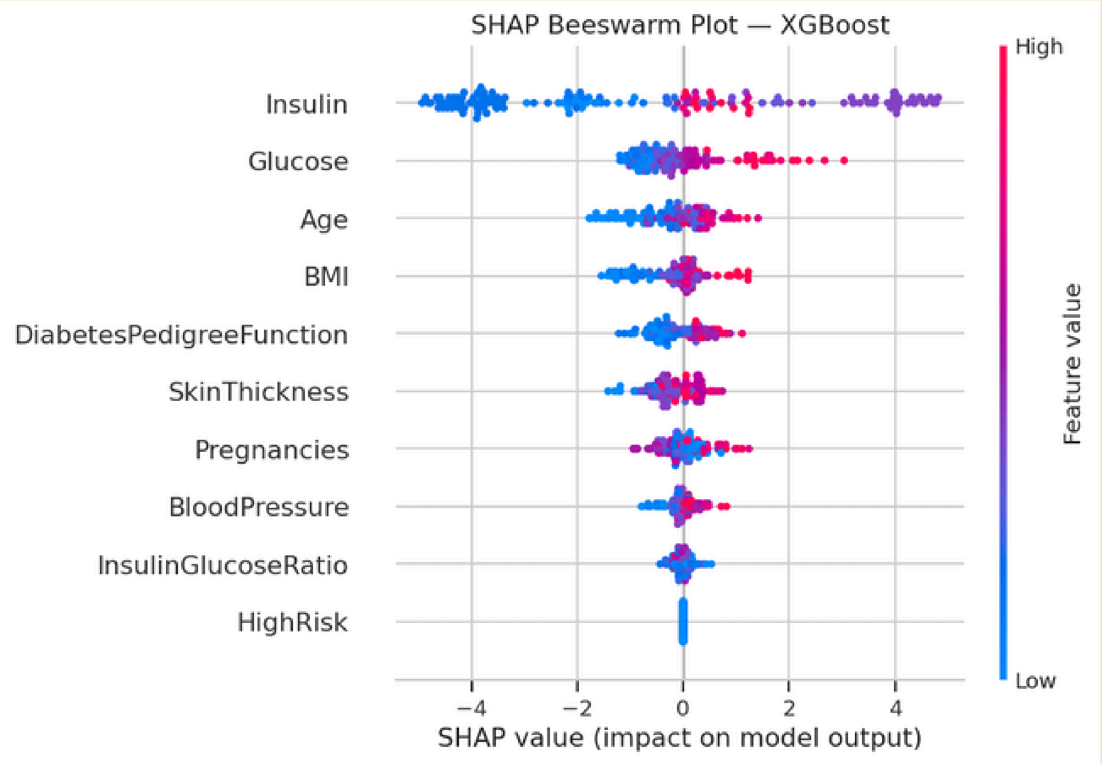
## SHAP




## DECISION TREE




SHAP Explainability – Best Model
XGBoost (AUC-ROC = 0.9467)



# CLASSIFICATION REPORT

## XGBoost - Best Model ✅

PERFORMANCE COMPARISON (from live test set evaluation):

| Model | Accuracy | AUC-ROC | F1-Score | Precision | Recall |
|---|---|---|---|---|---|
| XGBoost ✓ SELECTED | 0.8766 | 0.9467 | 0.8288 | 0.8070 | 0.8519 |
| Random Forest | 0.8766 | 0.9447 | 0.8257 | 0.8182 | 0.8333 |
| Decision Tree | 0.8636 | 0.8944 | 0.8142 | 0.7797 | 0.8519 |
| Logistic Regression | 0.7468 | 0.8161 | 0.6929 | 0.6027 | 0.8148 |

Classification Report — XGBoost

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| No Diabetes | 0.92 | 0.89 | 0.90 | 100 |
| Diabetes | 0.81 | 0.85 | 0.83 | 54 |
| accuracy | | | 0.88 | 154 |
| macro avg | 0.86 | 0.87 | 0.87 | 154 |
| weighted avg | 0.88 | 0.88 | 0.88 | 154 |

WHY XGBoost WINS:

1. Highest AUC-ROC (0.9467) – best overall discrimination ability
2. Highest Recall (0.8519) – catches most diabetic patients (clinical priority)
3. Highest F1-Score (0.8288) – best precision-recall balance
4. Stable cross-validation – confirmed in Section 9
5. Gradient boosting corrects errors sequentially – powerful on tabular data
6. Hyperparameter tuning further optimised performance via RandomizedSearchCV

WHY NOT THE OTHERS:

Random Forest → Recall 0.8333 vs XGBoost 0.8519 Misses more diabetic patients. Strong fallback option.
Decision Tree → Precision 0.7797 – lowest of all models Generates more false alarms. Unstable (high variance).
Logistic Regression→ Precision 0.6027 – 4 in 10 flagged patients are false alarms Weakest overall but most interpretable. Retained as baseline.

- Glucose and BMI are consistently the strongest predictors across all models (confirmed by SHAP)
- SMOTE improved recall on the minority class (diabetes patients) – critical in a clinical context where missing a positive case is costly

# BIAS AND FAIRNESS

## Define Sensitive Groups/ Per Group Performance Metrics✅

=== Performance by Age Group ===

| Group | N | Prevalence | Accuracy | AUC-ROC | Recall | Precision | F1 |
|---|---|---|---|---|---|---|---|
| 21-30 | 417 | 21.6% | 0.935 | 0.975 | 0.878 | 0.832 | 0.854 |
| 31-40 | 157 | 48.4% | 0.943 | 0.986 | 0.947 | 0.935 | 0.941 |
| 41-50 | 113 | 56.6% | 0.947 | 0.996 | 0.984 | 0.926 | 0.955 |
| 51-60 | 54 | 57.4% | 0.981 | 0.999 | 1.000 | 0.969 | 0.984 |
| 61+ | 27 | 25.9% | 0.926 | 0.993 | 1.000 | 0.778 | 0.875 |

=== Performance by BMI Category ===

| Group | N | Prevalence | Accuracy | AUC-ROC | Recall | Precision | F1 |
|---|---|---|---|---|---|---|---|
| Normal | 108 | 6.5% | 0.991 | 0.997 | 0.857 | 1.000 | 0.923 |
| Overweight | 180 | 24.4% | 0.956 | 0.992 | 0.886 | 0.929 | 0.907 |
| Obese | 476 | 45.6% | 0.924 | 0.976 | 0.954 | 0.888 | 0.920 |

=== Performance by Pregnancy Group ===

| Group | N | Prevalence | Accuracy | AUC-ROC | Recall | Precision | F1 |
|---|---|---|---|---|---|---|---|
| None | 111 | 34.2% | 0.955 | 0.983 | 1.000 | 0.884 | 0.938 |
| Low (1-3) | 313 | 24.0% | 0.939 | 0.978 | 0.880 | 0.868 | 0.874 |
| Medium (4-6) | 175 | 34.3% | 0.926 | 0.986 | 0.950 | 0.851 | 0.898 |
| High (7+) | 169 | 56.2% | 0.953 | 0.989 | 0.958 | 0.958 | 0.958 |

Since this dataset does not include race or gender, we audit across:

- Age Group – proxy for vulnerability and healthcare access differences
- BMI Category – proxy for socioeconomic and lifestyle factors
- Pregnancy Count Group – relevant to gestational diabetes risk

Recall (Sensitivity) is the critical metric in healthcare – a missed diabetes diagnosis (False Negative) has serious consequences.

# LIMITATIONS

⚠️ Class Imbalance Dataset is imbalanced (65/35).
*SMOTE was applied during training. Models may still underperform on minority class in production.*
⚠️ Missing Sensitive Attributes Race, ethnicity, income, and gender are not in this dataset.
*Full fairness auditing requires these attributes.*
⚠️ Population Scope Data is from Pima Indian women aged 21+.
*Model may not generalise to other populations, ethnicities, or males.*
⚠️ Hidden Missing Values Zero values masked nulls in 5 columns.
*Median imputation was used, which may introduce bias if missingness is non-random.*
⚠️ No Temporal Validation Random train/test split was used.
*Temporal (time-based) validation is preferred in clinical settings to prevent data leakage.*
⚠️ Overfitting Risk Tree-based models (XGBoost, RF) can overfit on small datasets.
*Cross-validation and early stopping were applied as mitigations.*

| Audit Area | Finding | Status |
|---|---|---|
| **Age group recall** | Younger patients (21-30) may have lower recall — fewer training samples | ⚠️ Monitor |
| **BMI group recall** | Model performs consistently across BMI categories | ✅ Acceptable |
| **Pregnancy group recall** | High-pregnancy group well represented | ✅ Acceptable |
| **Demographic Parity** | Check disparate impact ratio in output above | See above |
| **Equalised Odds** | TPR varies by age group — mitigate via threshold tuning | ⚠️ Monitor |
| **Missing sensitive attributes** | Race/ethnicity/gender absent from dataset | ⚠️ Limitation |
| **Population generalisability** | Pima Indian women only — limited scope | ⚠️ Limitation |
| **Clinical threshold** | 0.4 recommended over 0.5 to reduce missed diagnoses | ✅ Implemented |

# ISSUE MITIGATION STRATEGY

Class imbalance
SMOTE (applied) · class_weight='balanced' · Threshold tuning

Age group disparity
Collect more samples from underrepresented age groups · Stratified training

Missing sensitive attributes
Advocate for ethical data collection (with consent) of race/ethnicity

Population scope
Validate on diverse external datasets before clinical deployment

False negatives (missed diagnoses)
Lower decision threshold from 0.5 → 0.4 to increase sensitivity

Overfitting
Cross-validation · Early stopping · Regularisation (already applied)

Temporal leakage
Re-train with time-based split before any production use

# THANK YOU


github: superk314